2. 68.26% of the area under the standard normal curve lies between z = –1 and Z =1

3. 95.44% of the area lies between Z = –2 and Z = 2

4. 99.74% of the area lies between Z = –3 and Z = 3

Example 7.21

What is the probability that a standard normal variate Z will be

(i) greater than 1.09

(ii) less than –1.65

(iii) lying between –1.00 and 1.96

(iv) lying between 1.25 and 2.75

Solution :

(i) greater than 1.09

The total area under the curve is equal to 1 , so that the total area to the right Z = 0 is 0.5 (since the curve is symmetrical). The area between Z = 0 and 1.09 (from tables) is 0.3621

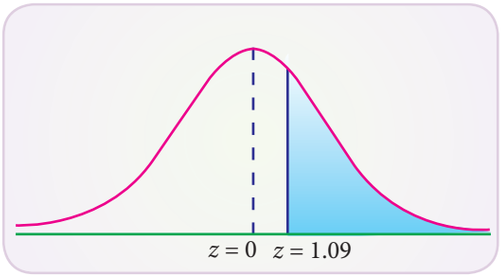


Fig 7.4

P(Z > 1.09) = 0.5000 – 0.3621 = 0.1379

The shaded area to the right of Z = 1.09 is the probability that Z will be greater than 1.09

(ii) less than –1.65

The area between –1.65 and 0 is the same as area between 0 and 1.65. In the table the area between zero and 1.65 is 0.4505 (from the table). Since the area to the left of zero is 0.5, P(Z< 1.65) = 0.5000 – 0.4505 = 0.0495.

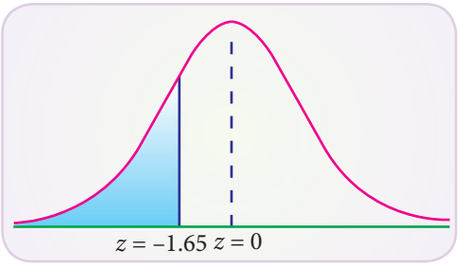


Fig 7.5

(iii) lying between –1.00 and 1.96

The probability that the random variable Z in between –1.00 and 1.96 is found by adding the corresponding areas :

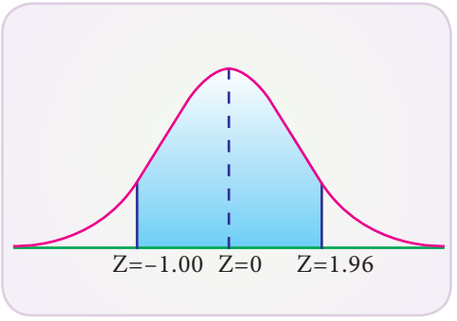


Fig 7.6

Area between –1.00 and 1.96 = area between (–1.00 and 0) + area betwn (0 and 1.96)

$$P(-1.00 < Z < 1.96) = P(-1.00 < Z < 0) + P(0 < Z < 1.96)$$

$$= 0.3413 + 0.4750 \text{ (by tables)}$$

$$= 0.8163$$

(iv) lying between 1.25 and 2.75

Area between $Z = 1.25$ and 2.75
= area between ($z = 0$ and $z = 2.75$)
– area between (z=0 and $z = 1.25$)

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